Tugas Praktikum Sesi 6
Pemrograman Mobile



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PROGRAM STUDI TEKNIK INFORMATIKA

FAKULTAS ILMU KOMPUTER

UNIVERSITAS ESA UNGGUL

TANGERANG

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1. Langkah Praktikum

a. Tema Terang/Gelap dan Warna Tema

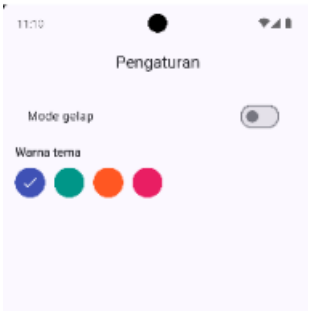
Ganti seluruh isi lib/main.dart. Dua ValueNotifier global menyimpan pilihan tema; ListenableBuilder membangun ulang MaterialApp saat salah satunya berubah.

```dart
import 'package:flutter/material.dart';
// Dua "penyimpan pilihan" global. Saat .value berubah, pendengarnya diberi tahu.
final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
  Colors.indigo,
  Colors.teal,
  Colors.deepOrange,
  Colors.pink,
];
// Fungsi pembuat tema: cukup 1 warna "seed", Material 3 membuat palet lengkapnya.
ThemeData buatTema(Color seed, Brightness brightness) {
  return ThemeData(
    useMaterial3: true,
    colorSchemeSeed: seed,
    brightness: brightness,
    appBarTheme: const AppBarTheme(centerTitle: true),
  );
}

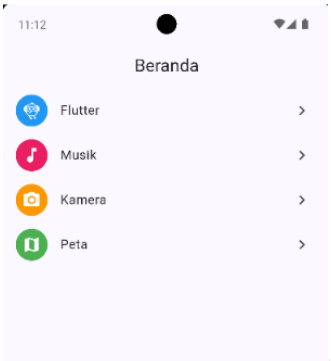
void main() => runApp(const MyApp());
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return ListenableBuilder(
      listenable: Listenable.merge([themeMode, seedColor]),
      builder: (context, _) {
        return MaterialApp(
          title: 'Praktikum 6',
          debugShowCheckedModeBanner: false,
          theme: buatTema(seedColor.value, Brightness.light),
          darkTheme: buatTema(seedColor.value, Brightness.dark),
          themeMode: themeMode.value,
          home: Scaffold(
            appBar: AppBar(title: const Text('Tema')),
            body: const PengaturanTab(),
          ),
        );
      },
    );
  }
}

class PengaturanTab extends StatelessWidget {
  const PengaturanTab({super.key});
  @override
  Widget build(BuildContext context) {
    return ListenableBuilder(
      listenable: Listenable.merge([themeMode, seedColor]),
      builder: (context, _) {
        return ListView(
          padding: const EdgeInsets.all(16),
          children: [
            SwitchListTile(
              title: const Text('Mode gelap'),
              value: themeMode.value == ThemeMode.dark,
              onChanged: (v) {
                themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
              },
            ),
            const SizedBox(height: 8),
            Text(
              'Warna tema',
              style: Theme.of(context).textTheme.titleMedium,
            ),
            const SizedBox(height: 8),
            Wrap(
              spacing: 12,
              children: [
                for (final c in pilihanWarna)
                  GestureDetector(
                    onTap: () => seedColor.value = c,
                    child: CircleAvatar(
                      backgroundColor: c,
                      child: seedColor.value == c
                          ? const Icon(Icons.check, color: Colors.white)
                          : null,
                    ),
                  ),
              ],
            ),
          ],
        );
      },
    );
  }
}
```



b. Named Routes, Argumen, dan Hero

Tambahkan model data dan halaman di bawah PengaturanTab:




```dart
// Model data: satu objek = satu item di daftar
class Item {
  final String nama;
  final IconData ikon;
  final Color warna;
  const Item(this.nama, this.ikon, this.warna);
}
const daftarItem = [
  Item('Flutter', Icons.flutter_dash, Colors.blue),
  Item('Musik', Icons.music_note, Colors.pink),
  Item('Kamera', Icons.camera_alt, Colors.orange),
  Item('Peta', Icons.map, Colors.green),
];

// Halaman daftar
class BerandaTab extends StatelessWidget {
  const BerandaTab({super.key});
  @override
  Widget build(BuildContext context) {
    return ListView.builder(
      itemCount: daftarItem.length,
      itemBuilder: (context, i) {
        final item = daftarItem[i];
        return ListTile(
          leading: Hero(
            tag: 'ikon-${item.nama}',
            child: CircleAvatar(
              backgroundColor: item.warna,
              child: Icon(item.ikon, color: Colors.white),
            ),
          ),
          title: Text(item.nama),
          trailing: const Icon(Icons.chevron_right),
          onTap: () {
            Navigator.pushNamed(context, '/detail', arguments: item);
          },
        );
      },
    );
  }
}
```

```dart
// Halaman detail
class DetailPage extends StatelessWidget {
  final Item item;
  const DetailPage({super.key, required this.item});
  @override
  Widget build(BuildContext context) {
    final tema = Theme.of(context);
    return Scaffold(
      appBar: AppBar(title: Text(item.nama)),
      body: Center(
        child: Column(
          mainAxisSize: MainAxisSize.min,
          children: [
            Hero(
              tag: 'ikon-${item.nama}',
              child: CircleAvatar(
                radius: 64,
                backgroundColor: item.warna,
                child: Icon(item.ikon, size: 64, color: Colors.white),
              ),
            ),
            const SizedBox(height: 16),
            Text(item.nama, style: tema.textTheme.headlineMedium),
            Text('Detail untuk ${item.nama}', style: tema.textTheme.bodyMedium),
          ],
        ),
      ),
    );
  }
}
```

Lalu pada MyApp, ganti baris home: (Scaffold Tema) dengan daftar rute berikut. Rute '/' didaftarkan di routes; rute yang membawa argumen ditangani onGenerateRoute

```
themeMode: themeMode.value,
initialRoute: '/',
routes: {
  '/': (_) => Scaffold(
    appBar: AppBar(title: const Text('Beranda')),
    body: const BerandaTab(),
  ), Scaffold
},
```

c. Animasi

Tambahkan halaman AnimasiTab yang berisi tiga demo: animasi implisit, AnimatedSwitcher, dan animasi eksplisit. Karena AnimationController membutuhkan vsync, state class memakai SingleTickerProviderStateMixin.

```
class AnimasiTab extends StatefulWidget {
  const AnimasiTab({super.key});

  @override
  State<AnimasiTab> createState() => _AnimasiTabState();
}

class _AnimasiTabState extends State<AnimasiTab>
    with SingleTickerProviderStateMixin {
  bool _besar = false;
  int _hitung = 0;
  late final AnimationController _putar;

  @override
  void initState() {
    super.initState();
    _putar = AnimationController(
      vsync: this,
      duration: const Duration(seconds: 2),
    )..repeat(); AnimationController
  }

  @override
  void dispose() {
    _putar.dispose();
    super.dispose();
  }

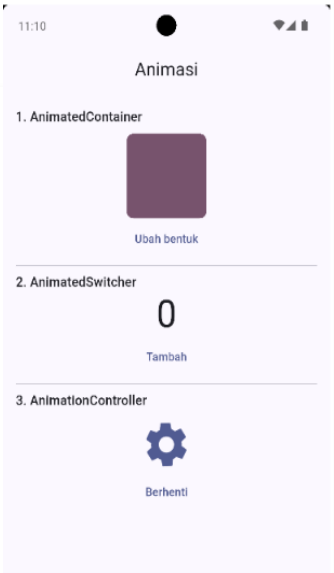
  @override
  Widget build(BuildContext context) {
    final tema = Theme.of(context);
    final warna = tema.colorScheme;
```

```
return ListView(
  padding: const EdgeInsets.all(16),
  children: [
    // ---------- 1. Animasi implisit ----------
    Text('1. AnimatedContainer', style: tema.textTheme.titleMedium),
    const SizedBox(height: 8),
    Center(
      child: AnimatedContainer(
        duration: const Duration(milliseconds: 500),
        curve: Curves.easeInOut,
        width: _besar ? 200 : 100,
        height: 100,
        decoration: BoxDecoration(
          color: _besar ? warna.primary : warna.tertiary,
          borderRadius: BorderRadius.circular(_besar ? 50 : 8),
        ), BoxDecoration
      ), AnimatedContainer
    ), Center
    TextButton(
      onPressed: () => setState(() => _besar = !_besar),
      child: const Text('Ubah bentuk'),
    ), TextButton
    const Divider(),
```

```
// ---------- 2. AnimatedSwitcher ----------
Text('2. AnimatedSwitcher', style: tema.textTheme.titleMedium),
Center(
  child: AnimatedSwitcher(
    duration: const Duration(milliseconds: 300),
    transitionBuilder: (child, anim) {
      return ScaleTransition(scale: anim, child: child);
    },
    child: Text(
      '$_hitung',
      key: ValueKey(_hitung),
      style: tema.textTheme.displayMedium,
    ), Text
  ), AnimatedSwitcher
), Center
TextButton(
  onPressed: () => setState(() => _hitung++),
  child: const Text('Tambah'),
), TextButton
const Divider(),
```

```
// ---------- 3. Animasi eksplisit ----------
Text('3. AnimationController', style: tema.textTheme.titleMedium),
const SizedBox(height: 8),
Center(
  child: RotationTransition(
    turns: _putar,
    child: Icon(Icons.settings, size: 64, color: warna.primary),
  ), RotationTransition
), Center
TextButton(
  onPressed: () {
    if (_putar.isAnimating) {
      _putar.stop();
    } else {
      _putar.repeat();
    }
    setState(() {});
  },
  child: Text(_putar.isAnimating ? 'Berhenti' : 'Putar'),
), TextButton
],
); ListView
}
}
```

```
routes: {
  '/': (_) => Scaffold(
    appBar: AppBar(title: const Text('Animasi')),
    body: const AnimasiTab(),
  ), Scaffold
},
```



d. Navigasi Tab Bawah (Menyatukan Semuanya)

Tambahkan ShellPage, yang menampung tiga tab. IndexedStack menjaga state setiap tab tetap hidup saat berpindah. (Terakhir, ubah rute '/' pada MyApp menjadi)

```dart
class ShellPage extends StatefulWidget {
  const ShellPage({super.key});

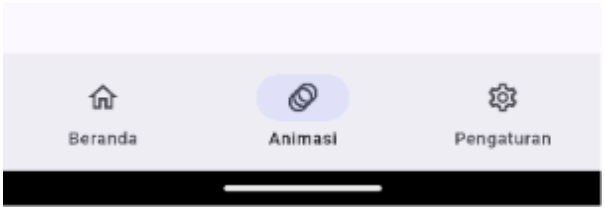
  @override
  State<ShellPage> createState() => _ShellPageState();
}

class _ShellPageState extends State<ShellPage> {
  int _index = 0;

  static const _halaman = [BerandaTab(), AnimasiTab(), PengaturanTab()];
  static const _judul = ['Beranda', 'Animasi', 'Pengaturan'];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(_judul[_index])),
      body: IndexedStack(index: _index, children: _halaman),
      bottomNavigationBar: NavigationBar(
        selectedIndex: _index,
        onDestinationSelected: (i) => setState(() => _index = i),
        destinations: const [
          NavigationDestination(
            icon: Icon(Icons.home_outlined),
            selectedIcon: Icon(Icons.home),
            label: 'Beranda',
          ), NavigationDestination
          NavigationDestination(
            icon: Icon(Icons.animation),
            label: 'Animasi',
          ), NavigationDestination
          NavigationDestination(
            icon: Icon(Icons.settings_outlined),
            selectedIcon: Icon(Icons.settings),
            label: 'Pengaturan',
          ), NavigationDestination
        ],
      ), NavigationBar
    ); Scaffold
  }
}
```

```dart
  initialRoute: '/',
  routes: {
    '/': (_) => const ShellPage(),
  },
); MaterialApp
```



2. Latihan Mandiri

1. Simpan mode gelap dan warna tema dengan shared_preferences (Pertemuan 5) dan muat saat aplikasi dimulai (WidgetsFlutterBinding.ensureInitialized() di main()).

```dart
// Latihan Mandiri no.1
Future<void> main() async {
  WidgetsFlutterBinding.ensureInitialized()

  // 1. Buka penyimpanan
  final prefs = await SharedPreferences.getInstance();

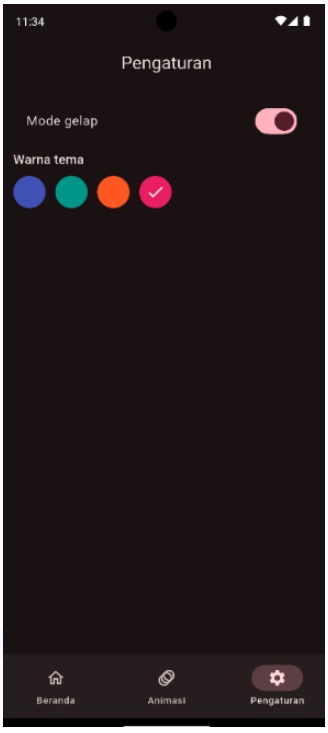
  // 2. Baca nilai tersimpan (kalau belum ada, pakai nilai awal setelah ??)
  final gelap = prefs.getBool('gelap') ?? false;
  final indeksWarna = prefs.getInt('warna') ?? 0;

  // 3. Masukkan ke ValueNotifier sebelum aplikasi tampil
  themeMode.value = gelap ? ThemeMode.dark : ThemeMode.light;
  seedColor.value =
  pilihanWarna[indeksWarna.clamp(0, pilihanWarna.length - 1)];

  // 4. Setiap nilai berubah, simpan otomatis
  themeMode.addListener(() {
    prefs.setBool('gelap', themeMode.value == ThemeMode.dark);
  });
  seedColor.addListener(() {
    prefs.setInt(
      'warna',
      pilihanWarna.indexWhere((c) => c == seedColor.value),
    );
  });

  runApp(const MyApp());
}
```

```dart
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';
```



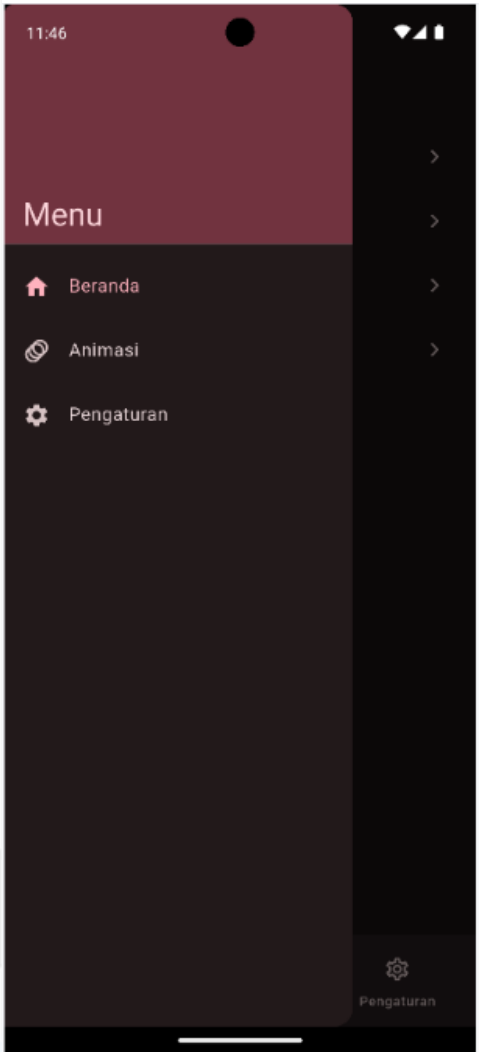
cttn : sudah disimpan otomatis (jadi jika di stop dan run Kembali, tidak akan terjadi default di awal)

2. Tambahkan Drawer pada ShellPage berisi menu untuk berpindah tab.

```
static const _ikon = [Icons.home, Icons.animation, Icons.settings];

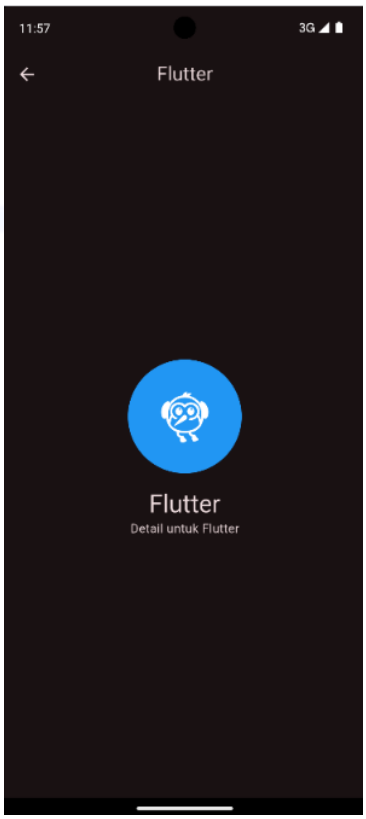
@override
Widget build(BuildContext context) {
  final tema = Theme.of(context);
  final warna = tema.colorScheme;

  return Scaffold(
    appBar: AppBar(title: Text(_judul[_index])),
    // ---------- Latihan Mandiri no.2 : Drawer ----------
    drawer: Drawer(
      child: ListView(
        padding: EdgeInsets.zero,
        children: [
          DrawerHeader(
            decoration: BoxDecoration(color: warna.primaryContainer),
            child: Align(
              alignment: Alignment.bottomLeft,
              child: Text(
                'Menu',
                style: tema.textTheme.headlineMedium?.copyWith(
                  color: warna.onPrimaryContainer,
                ),
              ), Text
            ), Align
          ), DrawerHeader
          for (int i = 0; i < _judul.length; i++)
            ListTile(
              leading: Icon(_ikon[i]),
              title: Text(_judul[i]),
              selected: _index == i,
              onTap: () {
                setState(() => _index = i);
                Navigator.pop(context); // tutup drawer
              },
            ), ListTile
        ],
      ), ListView
    ), Drawer
```



3. Ganti transisi halaman detail dengan PageRouteBuilder yang memakai FadeTransition (di dalam onGenerateRoute).

```
theme: buatTema(seedColor.value, Brightness.light),
darkTheme: buatTema(seedColor.value, Brightness.dark),
themeMode: themeMode.value,
initialRoute: '/',
routes: {
  '/': (_) => const ShellPage(),
},
// ---------- (Latihan Mandiri no. 3: fade) ----------
onGenerateRoute: (settings) {
  if (settings.name == '/detail') {
    final item = settings.arguments as Item;
    return PageRouteBuilder(
      settings: settings,
      transitionDuration: const Duration(milliseconds: 400),
      reverseTransitionDuration: const Duration(milliseconds: 300),
      pageBuilder: (context, animation, secondaryAnimation) {
        return DetailPage(item: item);
      },
      transitionsBuilder:
          (context, animation, secondaryAnimation, child) {
        return FadeTransition(opacity: animation, child: child);
      },
    ); PageRouteBuilder
  }
  return null;
},
// ------------------------------------------------------
); MaterialApp
```



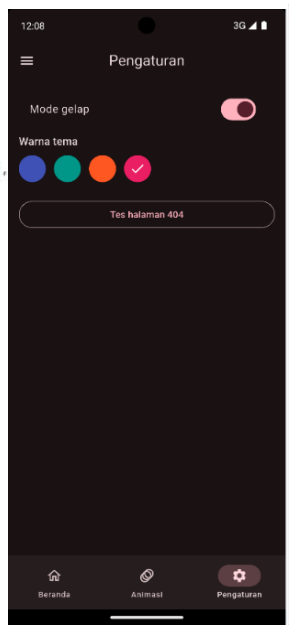
4. Tambahkan onUnknownRoute yang menampilkan halaman "404 - Halaman tidak ditemukan".

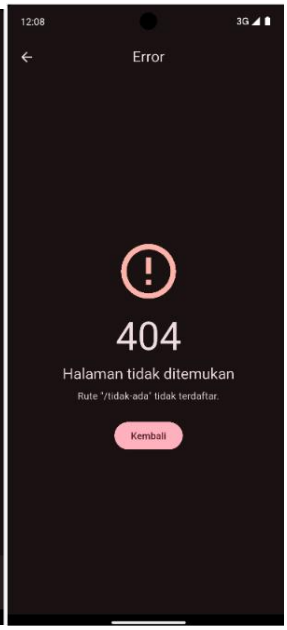
```
------- Latihan Mandiri no. 4 -------------
HalamanTidakDitemukan extends StatelessWidget {
al String? namaRute;
st HalamanTidakDitemukan({super.key, this.namaRute});

erride
get build(BuildContext context) {
inal tema = Theme.of(context);
inal warna = tema.colorScheme;

eturn Scaffold(
 appBar: AppBar(title: const Text('Error')),
 body: Center(
   child: Padding(
     padding: const EdgeInsets.all(24),
     child: Column(
       mainAxisSize: MainAxisSize.min,
       children: [
         Icon(Icons.error_outline, size: 96, color: warna.error),
         const SizedBox(height: 16),
         Text('404', style: tema.textTheme.displayLarge),
         const SizedBox(height: 8),
         Text(
           'Halaman tidak ditemukan',
           style: tema.textTheme.titleLarge,
         ), Text
         const SizedBox(height: 8),
         Text(
           'Rute "$namaRute" tidak terdaftar.',
           style: tema.textTheme.bodyMedium,
           textAlign: TextAlign.center,
         ), Text
         const SizedBox(height: 24),
         FilledButton(
           onPressed: () => Navigator.pop(context),
           child: const Text('Kembali'),
         ), FilledButton
       ],
     ), Column
```

```
// ------------------(tambahan LM no.4)-------------------------
  onUnknownRoute: (settings) {
    return MaterialPageRoute(
      settings: settings,
      builder: (_) => HalamanTidakDitemukan(namaRute: settings.name),
    ); MaterialPageRoute
  },
); MaterialApp
, ListenableBuilder
```

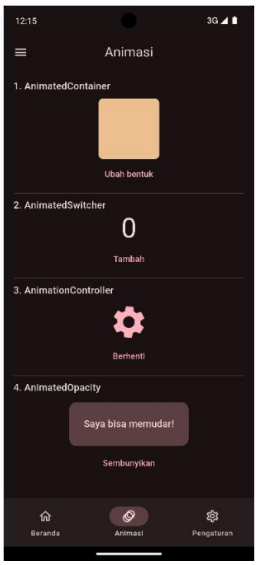




5. Tambahkan demo keempat di AnimasiTab: AnimatedOpacity yang memudarkan dan memunculkan sebuah kartu saat tombol ditekan

```
    // ---------- 4. AnimatedOpacity -------(Latihan Mandiri no.5)
    Text('4. AnimatedOpacity', style: tema.textTheme.titleMedium),
    const SizedBox(height: 8),
    Center(
      child: AnimatedOpacity(
        opacity: _tampil ? 1.0 : 0.0,
        duration: const Duration(milliseconds: 600),
        curve: Curves.easeInOut,
        child: Card(
          color: warna.secondaryContainer,
          child: Padding(
            padding: const EdgeInsets.all(24),
            child: Text(
              'Saya bisa memudar!',
              style: tema.textTheme.titleMedium?.copyWith(
                color: warna.onSecondaryContainer,
              ),
            ), Text
          ), Padding
        ), Card
      ), AnimatedOpacity
    ), Center
    TextButton(
      onPressed: () => setState(() => _tampil = !_tampil),
      child: Text(_tampil ? 'Sembunyikan' : 'Tampilkan'),
    ), TextButton
  ],
); ListView
  }
}
```

```
class _AnimasiTabState extends State<AnimasiTab>
    with SingleTickerProviderStateMixin {
  bool _besar = false;
  int _hitung = 0;
  bool _tampil = true; // tambah 1 untuk animasi
  late final AnimationController _putar;
```



3. Tugas

Buat aplikasi Galeri/Katalog Interaktif dengan tema bebas (misalnya profil diri, katalog produk, atau galeri tempat wisata) dengan ketentuan:

- Minimal 3 tab memakai NavigationBar (misalnya Beranda, Favorit/Katalog, Pengaturan).
- Named routes dengan argumen untuk halaman detail, dan halaman detail memakai Hero.
- Minimal dua jenis animasi: satu implisit dan satu eksplisit (AnimationController).
- Pengaturan tema terang/gelap dan minimal 3 pilihan warna, tersimpan setelah aplikasi ditutup.
- Warna dan gaya teks mengambil dari Theme.of(context) (hindari warna hard-coded pada teks dan latar).
- Kumpulkan: tangkapan layar (mode terang, mode gelap, halaman detail) dan berkas kode (atau tautan repositori). Aplikasi ini akan menjadi bahan awal untuk mini proyek di Pertemuan 7

Hasil Code di Folder tugas_6 : membuat aplikasi galeri wisata

```dart
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

// ================= TEMA =================
final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);
final favorit = ValueNotifier<Set<String>>({});

const pilihanWarna = [
  Colors.indigo,
  Colors.teal,
  Colors.deepOrange,
  Colors.pink,
];

void toggleFavorit(String id) {
  final baru = {...favorit.value};   // salin set lama
  if (!baru.remove(id)) baru.add(id); // sudah ada → hapus, belum → tambah
  favorit.value = baru;               // set baru → pendengar diberi tahu
}

ThemeData buatTema(Color seed, Brightness brightness) {
  return ThemeData(
    useMaterial3: true,
    colorSchemeSeed: seed,
    brightness: brightness,
    appBarTheme: const AppBarTheme(centerTitle: true),
  );
}
```

```dart
// ================= MAIN =================
Future<void> main() async {
  WidgetsFlutterBinding.ensureInitialized();

  final prefs = await SharedPreferences.getInstance();
  final gelap = prefs.getBool('gelap') ?? false;
  final indeksWarna = prefs.getInt('warna') ?? 0;

  themeMode.value = gelap ? ThemeMode.dark : ThemeMode.light;
  seedColor.value =
  pilihanWarna[indeksWarna.clamp(0, pilihanWarna.length - 1)];
  favorit.value = (prefs.getStringList('favorit') ?? []).toSet();

  themeMode.addListener(() {
    prefs.setBool('gelap', themeMode.value == ThemeMode.dark);
  });
  seedColor.addListener(() {
    prefs.setInt(
      'warna',
      pilihanWarna.indexWhere((c) => c == seedColor.value),
    );
  });
  favorit.addListener(() {
    prefs.setStringList('favorit', favorit.value.toList());
  });

  runApp(const MyApp());
}
```

```dart
// ================= DATA =================
class Tempat {
  final String id;
  final String nama;
  final String lokasi;
  final String deskripsi;
  final IconData ikon;
  final Color warna;
  final double rating;

  const Tempat({
    required this.id,
    required this.nama,
    required this.lokasi,
    required this.deskripsi,
    required this.ikon,
    required this.warna,
    required this.rating,
  });
}

const daftarTempat = [
  Tempat(
    id: 'bromo',
    nama: 'Gunung Bromo',
    lokasi: 'Jawa Timur',
    deskripsi:
    'Gunung berapi aktif dengan lautan pasir dan pemandangan matahari '
        'terbit yang terkenal. Cocok didaki dini hari.',
    ikon: Icons.landscape,
    warna: Colors.orange,
    rating: 4.8,
  ),
  Tempat(
    id: 'borobudur',
    nama: 'Candi Borobudur',
    lokasi: 'Jawa Tengah',
    deskripsi:
    'Candi Buddha terbesar di dunia dari abad ke-9, warisan dunia '
        'UNESCO dengan ribuan relief dan stupa.',
    ikon: Icons.account_balance,
    warna: Colors.brown,
```

```dart
// ================= APLIKASI =================
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return ListenableBuilder(
      listenable: Listenable.merge([themeMode, seedColor]),
      builder: (context, _) {
        return MaterialApp(
          title: 'Galeri Wisata',
          debugShowCheckedModeBanner: false,
          theme: buatTema(seedColor.value, Brightness.light),
          darkTheme: buatTema(seedColor.value, Brightness.dark),
          themeMode: themeMode.value,
          initialRoute: '/',
          routes: {
            '/': (_) => const ShellPage(),
          },
          onGenerateRoute: (settings) {
            if (settings.name == '/detail') {
              final tempat = settings.arguments as Tempat;
              return PageRouteBuilder(
                settings: settings,
                transitionDuration: const Duration(milliseconds: 400),
                reverseTransitionDuration: const Duration(milliseconds: 300),
                pageBuilder: (context, animation, secondaryAnimation) {
                  return DetailPage(tempat: tempat);
                },
                transitionsBuilder:
                    (context, animation, secondaryAnimation, child) {
                  return FadeTransition(opacity: animation, child: child);
                },
              );
            }
            return null;
          },
          onUnknownRoute: (settings) {
            return MaterialPageRoute(
              settings: settings,
              builder: (_) => HalamanTidakDitemukan(namaRute: settings.name),
            );
          },
        );
      },
    );
  }
}
```

```dart
// ================= SHELL (3 TAB) =================
class ShellPage extends StatefulWidget {
  const ShellPage({super.key});

  @override
  State<ShellPage> createState() => _ShellPageState();
}

class _ShellPageState extends State<ShellPage> {
  int _index = 0;

  static const _halaman = [GaleriTab(), FavoritTab(), PengaturanTab()];
  static const _judul = ['Galeri Wisata', 'Favorit', 'Pengaturan'];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(_judul[_index])),
      body: IndexedStack(index: _index, children: _halaman),
      bottomNavigationBar: NavigationBar(
        selectedIndex: _index,
        onDestinationSelected: (i) => setState(() => _index = i),
        destinations: const [
          NavigationDestination(
            icon: Icon(Icons.photo_library_outlined),
            selectedIcon: Icon(Icons.photo_library),
            label: 'Galeri',
          ),
          NavigationDestination(
            icon: Icon(Icons.favorite_border),
            selectedIcon: Icon(Icons.favorite),
            label: 'Favorit',
          ),
          NavigationDestination(
            icon: Icon(Icons.settings_outlined),
            selectedIcon: Icon(Icons.settings),
            label: 'Pengaturan',
          ),
        ],
      ),
    );
  }
}
```

```dart
// ================= TAB 1: GALERI =================
const kisiTempat = SliverGridDelegateWithFixedCrossAxisCount(
  crossAxisCount: 2,
  crossAxisSpacing: 12,
  mainAxisSpacing: 12,
  childAspectRatio: 0.9,
);

class GaleriTab extends StatelessWidget {
  const GaleriTab({super.key});

  @override
  Widget build(BuildContext context) {
    return GridView.builder(
      padding: const EdgeInsets.all(16),
      gridDelegate: kisiTempat,
      itemCount: daftarTempat.length,
      itemBuilder: (context, i) => KartuTempat(tempat: daftarTempat[i]),
    );
  }
}

class KartuTempat extends StatelessWidget {
  final Tempat tempat;
  final bool pakaiHero;
  const KartuTempat({super.key, required this.tempat, this.pakaiHero = true});

  @override
  Widget build(BuildContext context) {
    final tema = Theme.of(context);

    final avatar = CircleAvatar(
      radius: 36,
      backgroundColor: tempat.warna,
      child: Icon(tempat.ikon, size: 36, color: Colors.white),
    );

    return Card(
      clipBehavior: Clip.antiAlias,
      child: Stack(
        fit: StackFit.expand,
        children: [
          InkWell(
            onTap: () {
              Navigator.pushNamed(context, '/detail', arguments: tempat);
            },
            child: Padding(
              padding: const EdgeInsets.all(12),
              child: Column(
                mainAxisAlignment: MainAxisAlignment.center,
                children: [
                  pakaiHero
                      ? Hero(tag: 'tempat-${tempat.id}', child: avatar)
                      : avatar,
                  const SizedBox(height: 12),
                  Text(
                    tempat.nama,
                    style: tema.textTheme.titleMedium,
                    textAlign: TextAlign.center,
                  ),
                  Text(tempat.lokasi, style: tema.textTheme.bodySmall),
                  const SizedBox(height: 4),
                  Row(
                    mainAxisSize: MainAxisSize.min,
                    children: [
                      Icon(Icons.star,
                          size: 16, color: tema.colorScheme.tertiary),
                      const SizedBox(width: 4),
                      Text('${tempat.rating}',
                          style: tema.textTheme.labelLarge),
                    ],
                  ),
                ],
              ),
            ),
          ),
          Positioned(
            top: 0,
            right: 0,
            child: TombolFavorit(id: tempat.id),
          ),
        ],
      ),
    );
  }
}

class TombolFavorit extends StatelessWidget {
  final String id;
  const TombolFavorit({super.key, required this.id});

  @override
  Widget build(BuildContext context) {
    final warna = Theme.of(context).colorScheme;

    return ValueListenableBuilder<Set<String>>(
      valueListenable: favorit,
      builder: (context, daftar, _) {
        final suka = daftar.contains(id);

        return IconButton(
          tooltip: suka ? 'Hapus dari favorit' : 'Tambah ke favorit',
          onPressed: () => toggleFavorit(id),
          icon: AnimatedSwitcher(
            duration: const Duration(milliseconds: 300),
            transitionBuilder: (child, anim) {
```

```dart
// ================= TAB 2: FAVORIT =================
class FavoritTab extends StatelessWidget {
  const FavoritTab({super.key});

  @override
  Widget build(BuildContext context) {
    final tema = Theme.of(context);

    return ValueListenableBuilder<Set<String>>(
      valueListenable: favorit,
      builder: (context, daftar, _) {
        final tempatFavorit =
        daftarTempat.where((t) => daftar.contains(t.id)).toList();

        if (tempatFavorit.isEmpty) {
          return Center(
            child: Column(
              mainAxisSize: MainAxisSize.min,
              children: [
                Icon(Icons.favorite_border,
                    size: 72, color: tema.colorScheme.outline),
                const SizedBox(height: 16),
                Text('Belum ada favorit', style: tema.textTheme.titleMedium),
                const SizedBox(height: 4),
                Text('Ketuk ikon hati pada sebuah tempat',
                    style: tema.textTheme.bodyMedium),
              ],
            ),
          );
        }

        return GridView.builder(
          padding: const EdgeInsets.all(16),
          gridDelegate: kisiTempat,
          itemCount: tempatFavorit.length,
          itemBuilder: (context, i) {
            return KartuTempat(tempat: tempatFavorit[i], pakaiHero: false);
          },
        );
      },
    );
  }
}
```

